

SHIVAM'S WORLD

Interactive Game-Based Portfolio

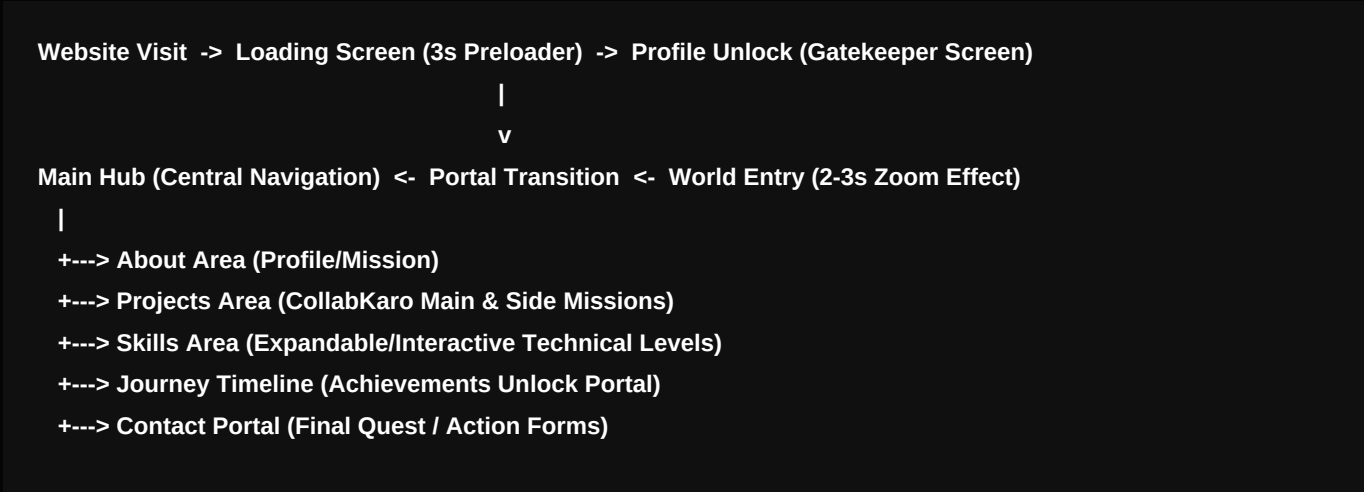
PRODUCT REQUIREMENTS DOCUMENT

Version:	1.0
Owner:	Shivam Singh
Date:	June 14, 2026
Status:	Approved / Ready for Build

1. Executive Summary & Objectives

Shivam's World is a fully interactive, game-inspired portfolio experience. Unlike traditional scrolling websites, visitors navigate between different scenes using clicks, animations, transition timings, sound effects, and immersive game-like interactions. The system is designed to mimic a high-fidelity game menu while preserving a crisp, professional identity optimized for recruiters, startup founders, brands, and collaborators.

2. User Journey Flow



3. Technical Architecture & Stack

- **Core Stack:** Next.js 15+ (App Router), React 19+, TypeScript
- **Styling & UI:** Tailwind CSS, custom CSS Variables (#050505 Dark theme only)
- **Animations:** Framer Motion (Transitions, Page Swaps), GSAP (Portal / Camera Zooms)
- **Audio Engine:** Howler.js (Preloading, Volume state, Sound on/off toggle)
- **Icons & Media:** Lucide React, optimized WebP / AVIF assets with lazy loading
- **Deployment:** Vercel & Git version control

4. Scene-by-Scene Functional Specifications

Scene Name	Core Purpose	Components	Interactions & Animations	Sound Trigger
Loading	Preload assets & timer	Logo, progress bar (0-100%)	Auto transition after 3s	Loading / End Chime
Profile Unlock	Gatekeep entrance	Portrait, Player Found text	Image hover scale 1.05; click zoom	Gate Unlock
World Entry	Portal zoom sequence	Portal animation, zoom text	Auto loads hub after 2-3s	Portal Ambient
Main Hub	Scene navigation	Scene selection cards	Hover border glow; click load scene	Hub Theme (Loop)
About	Bio & focus areas	Photo, mission, WebDev/AI c	Fade transition; click Back to Hub	Hub Theme (Muted)
Projects	Showcase work	Mission: CollabKaro, links	Click to view mission details	Mission Activation
Skills	List capabilities	Java, WebDev, React, DSA c	Expandable cards; slide reveal	Knowledge Unlock
Journey	Career timeline	Level checkpoints (1 to 8)	Click level to open detailed modal	Checkpoint chime
Achievements	Gamification log	Founder Mindset, Learner ca	Journey -> view achievements modal	Achievement Fanfare
Contact	Get in touch	Final Quest form, social links	Send message; social link hover	Portal Ambience

5. Audio & Sound System (Mandatory)

Audio acts as the emotional anchor of this portfolio experience. Powered by Howler.js, the system features global controls (Mute On/Off and Volume Slider) which are persisted in the user's browser local storage. Sounds are mapped to transitions and interactions as follows:

- **Ambient Loops:** Loading Preloader (drone), Main Hub Theme (low-tempo synth), Portal Ambience (Contact Scene).
- **Interactive Cues:** Hover UI (micro-beeps), Click UI (tactile tick), Mission Unlock (swoosh), Checkpoint (level note).
- **Achievement Chimes:** Fanfare notes when achievements are successfully unlocked or completed.

6. Design System & Theme Constraints

Token Name	Hex Value	Description / Usage Context
Background	#050505	Deepest black for canvas backdrop
Surface Panels	#101010	Container surface card backgrounds
Focus Cards	#151515	Active focal card elements
Borders	#242424	Subtle divider borders & visual separators
Primary Text	#FFFFFF	High-contrast titles and main instructions
Secondary Text	#A3A3A3	Subdued descriptive copy and metadata

[!] DESIGN RESTRICTION WARNING

No gradients, RGB color loops, or neon effects are permitted. The design must be kept clean, minimal and cinematic.

7. Animation Standards

Animations must feel cohesive, snappy, and organic. Standard parameters are as follows:

- Page/Scene Transition: 600ms duration (Fade, Zoom, and horizontal slide transitions)
- Easing Curve: cubic-bezier(0.25, 1, 0.5, 1) (Out Quart)
- Card Hover Transitions: Scale to 1.03 over 200ms
- Interactive Button Click: Scale down to 0.98 over 150ms for physical click feedback

8. Browser Local Storage Specifications

```
{
  "audioSettings": {
    "muted": false,
    "volume": 0.5
  },
  "exploration": {
    "visitedScenes": ["loading", "profile_unlock", "main_hub"],
    "completedLevels": ["level_1", "level_2"]
  },
  "achievements": {
    "unlocked": ["continuous_learner"]
  }
}
```

9. Performance, SEO & Roadmap

Performance Targets: Maintain Lighthouse scores of 90+ across all metrics. Load assets (audio, portrait illustrations) lazily, using Next.js Dynamic Imports. Optimized WebP/AVIF formats are mandatory.

Version 2.0 Roadmap (Gamification): Developer secret console (terminal keys), hidden easter eggs, and dynamic profile levels (XP counter).

Version 3.0 Roadmap (3D Immersive): Integration of React Three Fiber / Three.js 3D navigation desk and global visitor multiplayer visualizer.